

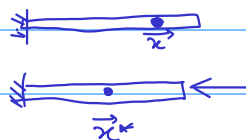
Displacement and Strain

- Rock Mechanics
 - Mechanics of particles & rigid bodies
- } Fundamental
Kinematic Variable
- Displacement.
- Displacement → Quantifies the change in position of a given particle of rock.

In typical Rock Mechanics Problem

- position of each particle
 ↳ Labelled w.r.t some Co-ordinate system.
 Called the initial state.
 position denoted by $\vec{x} = \langle x, y, z \rangle$
- Loads are then applied
 particle at position $\vec{x}$ is displaced to $\vec{x}^* = \langle x^*, y^*, z^* \rangle$
- The new Vector connecting $\vec{x}$ (original position) to $\vec{x}^*$ (final position) }
 ↓
 The Vector is denoted by $\vec{u} = \langle u, v, w \rangle$. Displacement at $\vec{x}$.

- To remain consistent with Sign Conventions of traction
 (+ve if traction points in -ve co-ordinate direction)

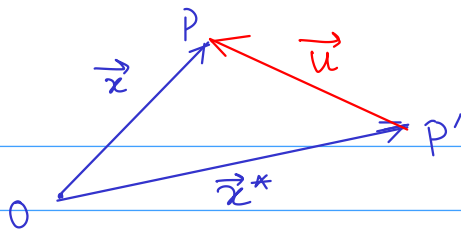


$$\vec{u} = \vec{x} - \vec{x}^*$$

eg:- Compression is +ve] traction.
 Shortening is +ve] Displacement

in general

$$\vec{x}^* = \vec{x} - \vec{u} \quad \text{ie; } \begin{aligned} x^* &= x - u \\ y^* &= y - v \\ z^* &= z - w \end{aligned}$$



Therefore, the displacement Vector $\vec{u}$

↓ Interpreted as

Vector that points from New position towards original position.

NB:- In general the displacement varies from point to point

$\therefore$ each Component $u, v, w \rightarrow$ Varies with all three position Co-ords.
(x, y, z)

Example:- The displacement field of a body is given by

$$\vec{u} = (x^2 + y)\hat{i} + (3 + z)\hat{j} + (x^2 + 2y)\hat{k}$$

What is the deformed position of a point originally at $(3, 1, -2)$?

Soln:- Displacement vector $\vec{u}$ at $(3, 1, -2)$

$$\vec{u} = (3^2 + 1)\hat{i} + \{3 + (-2)\}\hat{j} + (3^2 + 2 \times 1)\hat{k}$$

$$\vec{u} = 10\hat{i} + \hat{j} + 11\hat{k}$$

Initial position vector $\vec{r}$ at point P

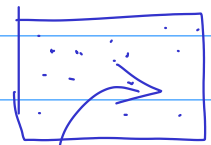
$$\vec{r} = 3\hat{i} + \hat{j} - 2\hat{k}$$

Now, Final position after deformation

$$\vec{r}^* = \vec{r} - \vec{u}$$

$$= (3 - 10)\hat{i} + (1 - 1)\hat{j} + (-2 + 11)\hat{k}$$

$$\vec{r}^* = -7\hat{i} + 9\hat{k}$$



$$\vec{u} = u\hat{i} + v\hat{j} + w\hat{k}$$

$$\begin{cases} u = f_1(x, y, z) \\ v = f_2(x, y, z) \\ w = f_3(x, y, z) \end{cases}$$

One of the objectives in Solving Rock Mechanics Problem.

- ↓
- Calculate the displacement Vector $\vec{u}$ at every point in rock mass given
 - applied traction
 - boundary stresses
 - body forces.

Requires a set of
Intermediate Quantities
(STRAINS)

- Stresses are directly related to Strain than to displacement
- Use of Strain rather than displacement analogous to using Stress $\leftrightarrow$ traction.

Mech Engrg / Theory of Mech of Solids
↓
Deals with
- Stress & Strain rather than traction & disp.

Rock Mech

- Stress / Strain and
- However $\rightarrow$ at times displacements are also important
eg:
 - closure of mine openings
 - well bore breakout
 - Subsidence etc.

Strain:- Strain is essentially a measure of relative displacement of nearby particles

rather than

a measure of their absolute displacement.

Concept of Strain (one dimensional context)

Initial Length of Bar $= L = \Delta x$.

A B Initial State
(x) (x + Δx)

New Length Bar $L^* = [(x + \Delta x) - u(x + \Delta x)] - [x - u(x)]$

A* B* (After application of stress)
x - u(x) (x + Δx) - u(x + Δx)

Note

Fractional Decrease in Length of bar $\left. \vphantom{\frac{L-L^*}{L}} \right\} = \frac{L - L^*}{L} = \frac{u(x + \Delta x) - u(x)}{\Delta x}$

Strain +ve $\rightarrow$ When bar shortens
-ve $\rightarrow$ When bar extends.

Strain at a point (One-Dimensional Case)

Limiting condition of above defn when $L \rightarrow 0$

$$\epsilon(x) = \lim_{L \rightarrow 0} \left(\frac{L - L^*}{L} \right) = \lim_{L \rightarrow 0} \frac{u(x + \Delta x) - u(x)}{\Delta x}$$

$\Rightarrow$ Strain is Spatial derivative of displacement.

- This type of Strain derived above is known as

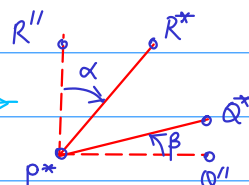
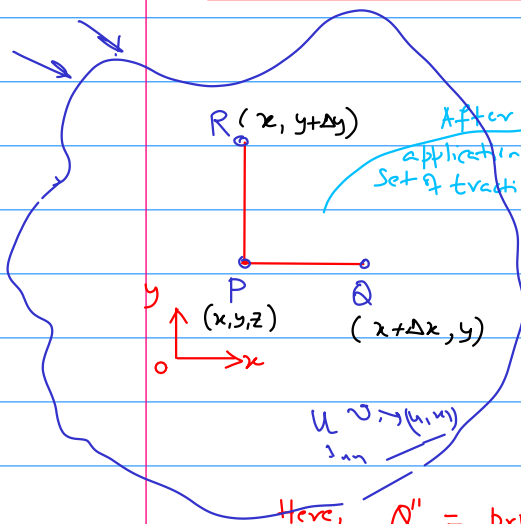
NORMAL STRAIN.

Can be generalised to 2 or 3-Dimensions.

- However, in higher dimensions we encounter other strains also
i.e., Shear Strains

$\rightarrow$ Measures the angular distortion rather than Stretching/contraction.

Infinitesimal Strain in Two Dimensions:



$$\begin{aligned} P^* &= P - u(P) = (x, y) - [u(x, y), v(x, y)] \\ &= [x - u(x, y), y - v(x, y)] \end{aligned} \quad \text{--- (1)}$$

$$\begin{aligned} Q^* &= Q - u(Q) = [x + \Delta x, y] - [u(x + \Delta x, y), v(x + \Delta x, y)] \\ &= \{x + \Delta x - u(x + \Delta x, y), y - v(x + \Delta x, y)\} \end{aligned} \quad \text{--- (2)}$$

$$\begin{aligned} R^* &= R - u(R) = (x, y + \Delta y) - [u(x, y + \Delta y), v(x, y + \Delta y)] \\ &= [x - u(x, y + \Delta y), y + \Delta y - v(x, y + \Delta y)] \end{aligned} \quad \text{--- (3)}$$

Here, Q'' = projection of Q^* onto x -axis passing through P^*
 R'' = projection of R^* onto y -axis passing through P^*

So, Q'' will have same x -component as Q^* & same y -component as P^* .

& Likewise for R''

Now, expressing each of the displacements appearing in P^*, Q^*, R^* as Taylor's Series & ignoring terms higher than the 1st order we get.

$$P^* = (x - u, y - v) \quad \text{--- (4)}$$

$$Q^* = \left(x + \Delta x - u - \frac{\partial u}{\partial x} \Delta x, y - v - \frac{\partial v}{\partial x} \Delta x \right) \quad \text{--- (5)}$$

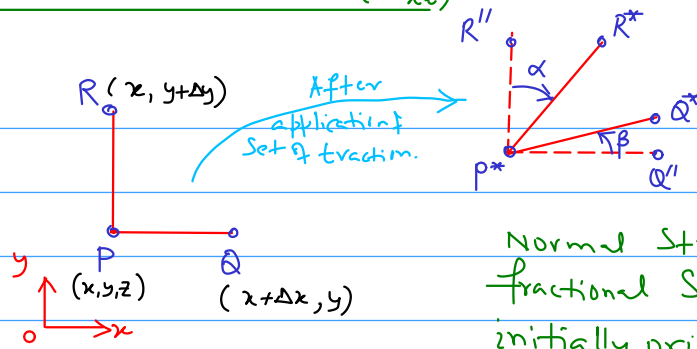
$$R^* = \left(x - u - u \frac{\partial u}{\partial y} \Delta y, y + \Delta y - v - \frac{\partial v}{\partial y} \Delta y \right) \quad \text{--- (6)}$$

$$Q'' = \left(x + \Delta x - u - \frac{\partial u}{\partial x} \Delta x, y - v \right) \quad \text{--- (7)}$$

$$R'' = \left(x - u, y + \Delta y - v - \frac{\partial v}{\partial y} \Delta y \right) \quad \text{--- (8)}$$

It is assumed that the Partial Derivatives are evaluated at point (x, y)

Normal Strain (ϵ_{xx})



Normal Strain ϵ_{xx} is defined as the fractional shortening of the line that is initially oriented along x -axis.

i.e. — ϵ_{xx} = Contraction undergone by PQ in limits $\Delta x \rightarrow 0$.

Initial length of PQ = Δx

Final length PQ = P^*Q^* .

$$\begin{aligned} \text{Now } |P^*Q^*|^2 &= |P^*Q''|^2 + |Q''Q^*|^2 \\ &= \left(\Delta x - \frac{\partial u}{\partial x} \Delta x\right)^2 + \left(\frac{\partial v}{\partial x} \Delta x\right)^2 \\ &= (\Delta x)^2 \left[1 - 2\frac{\partial u}{\partial x} + \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 \right] \quad \text{--- (9)} \end{aligned}$$

Assumptions in Theory of Small Strains: — partial derivatives of displacement are much smaller than unity.

$\therefore$ the strains are linear fun. of partial derivatives of displacement.

Widely use for: — Situations of Engg. interest in Rock Mechanics
e.g. — tunnels, mines, propagation of seismic waves etc.

Where as Theory of finite Strains $\rightarrow$ Geological Process

$\therefore$ Neglecting square of derivatives & using the fact
 $(1 - 2\delta)^{1/2} = 1 - \delta \quad \forall \delta \ll 1$.

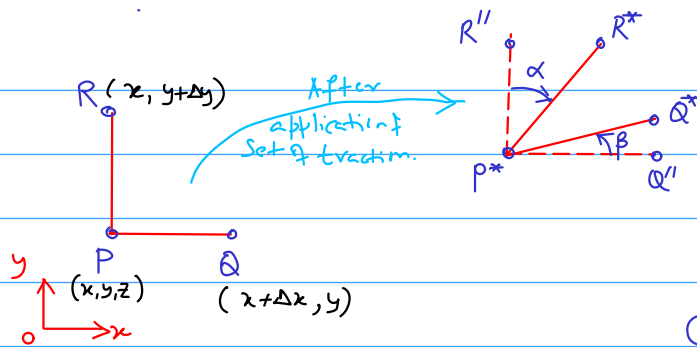
$$|P^*Q^*| = \Delta x \left(1 - \frac{\partial u}{\partial x} \right) \quad \text{--- (10)}$$

$\therefore$ Normal Strain in x -direction at point (x, y)

$$\begin{aligned} \epsilon_{xx} &= \lim_{\Delta x \rightarrow 0} \frac{|PQ| - |P^*Q^*|}{|PQ|} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\Delta x - \Delta x \left(1 - \frac{\partial u}{\partial x} \right)}{\Delta x} = \frac{\partial u}{\partial x} \quad \text{--- (11)} \end{aligned}$$

$$\parallel y. \quad \epsilon_{yy} = \lim_{\Delta y \rightarrow 0} \frac{\Delta y - \Delta y \left(1 - \frac{\partial v}{\partial y} \right)}{\Delta y} = \frac{\partial v}{\partial y} \quad \text{--- (12)}$$

Shear Strain in 2D

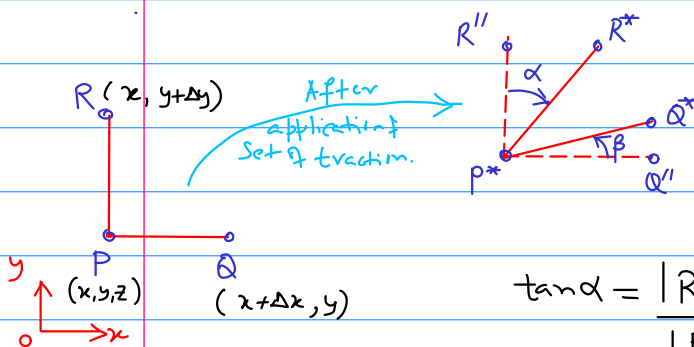


Initially:- P, Q & R form a right angle.

After deformation → Acute angle.

Change in the angle quantified as Shear strain.

Defn:- The shear strain (ϵ_{xy}) is defined as half of the increase in angle formed by two infinitesimal line segments that are initially parallel to the x & y -axes.



$$\epsilon_{xy} = \frac{1}{2} \lim_{\Delta x, \Delta y \rightarrow 0} (\angle R^*P^*Q^* - \angle RPQ) \quad \text{--- (13)}$$

$$\angle R^*P^*Q^* = 90^\circ - \alpha - \beta \quad \& \quad \angle RPQ = 90^\circ$$

$$\tan \alpha = \frac{|R^*R''|}{|P^*R''|} = \frac{-\frac{\partial u}{\partial y} \Delta y}{\Delta y (1 - \frac{\partial v}{\partial y})} = -\frac{\partial u}{\partial y} \quad \text{--- (14)}$$

NB:- ~~$\Delta y \cdot \frac{\partial v}{\partial y}$ ignored~~

$$\text{or, } \alpha = -\frac{\partial u}{\partial y} \quad \parallel \quad \beta = -\frac{\partial v}{\partial x}$$

$$(1 - \frac{\partial v}{\partial y})^{-1} = 1 + \frac{\partial v}{\partial y}$$

$$\text{So, } -\frac{\partial u}{\partial y} (1 + \frac{\partial v}{\partial y}) = -\frac{\partial u}{\partial y} - \frac{\partial u \partial v}{\partial y^2}$$

$$\therefore \epsilon_{xy} = \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \quad \text{--- (15)}$$

The pattern of above Eqn shows that

ϵ_{xy} = Average of

- (i) partial derivative of displacement in x -direction w.r.t y &
- (ii) ——— do ——— y -direction w.r.t x

$\therefore$ By this defn if ϵ_{yx} is defined it will necessarily be $= \epsilon_{xy}$.

The four strains are therefore thought to be components of strain matrix

$$\epsilon = \begin{bmatrix} \epsilon_{xx} & \epsilon_{xy} \\ \epsilon_{yx} & \epsilon_{yy} \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \\ \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) & \frac{\partial v}{\partial y} \end{bmatrix} \quad \text{--- (16)}$$

Note:- Engineering Shear Strain $\gamma_{xy} = 2\epsilon_{xy}$.

Some properties of Strain matrix.

- (1) Strain matrix is equal to the Symmetric part of displacement gradient $\nabla \vec{u}$

$$\epsilon = \text{Sym}(\nabla \vec{u}) = \frac{1}{2} \left\{ \nabla \vec{u} + (\nabla \vec{u})^T \right\} \quad \text{--- (17)}$$

$$\text{Where, } \nabla \vec{u} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix} \quad \text{--- (18)}$$

- (2) The displacement gradient and hence also the Strain is a 2nd order tensor.

└─> Under the rotation of co-ordinate axes
The components transform according to the law
 $\epsilon' = L \epsilon L^T$.

Proof:- Under rotation of co-ordinate axes

$$\vec{r}' = L \vec{r} \quad \text{We have } x' = x \cos \theta + y \sin \theta \text{ \& } y' = -x \sin \theta + y \cos \theta \quad \text{--- (19)}$$

$$\left. \begin{array}{l} \text{The inverse} \\ \text{Transformations} \end{array} \right\} x = x' \cos \theta - y' \sin \theta \text{ \& } y = x' \sin \theta + y' \cos \theta. \quad \text{--- (20)}$$

Since displacement is also a vector

$$\therefore u' = u \cos \theta + v \sin \theta \text{ \& } v' = -u \sin \theta + v \cos \theta \quad \text{--- (21)}$$

Now consider Component of Strain; eg: $\epsilon_{x'x'}$ & Use the chain rule.

$$\begin{aligned} \epsilon_{x'x'} &= \frac{\partial u'}{\partial x'} = \underbrace{\frac{\partial u'}{\partial u}}_c \underbrace{\frac{\partial u}{\partial x}}_c \underbrace{\frac{\partial x}{\partial x'}}_c + \underbrace{\frac{\partial u'}{\partial u}}_c \underbrace{\frac{\partial u}{\partial y}}_s \underbrace{\frac{\partial y}{\partial x'}}_s + \underbrace{\frac{\partial u'}{\partial v}}_s \underbrace{\frac{\partial v}{\partial x}}_c \underbrace{\frac{\partial x}{\partial x'}}_c + \underbrace{\frac{\partial u'}{\partial v}}_s \underbrace{\frac{\partial v}{\partial y}}_s \underbrace{\frac{\partial y}{\partial x'}}_s \\ &= \frac{\partial u}{\partial x} \cos^2 \theta + \frac{\partial u}{\partial y} \sin \theta \cos \theta + \frac{\partial v}{\partial x} \sin \theta \cos \theta + \frac{\partial v}{\partial y} \sin^2 \theta. \end{aligned}$$

$$\therefore \epsilon_{x'x'} = \epsilon_{xx} \cos^2 \theta + 2 \epsilon_{xy} \sin \theta \cos \theta + \epsilon_{yy} \sin^2 \theta \quad \text{--- (192)}$$

This is identical to Stress transformation law as in 2D for τ_{xx} . Similar analysis of other components reveal that

$$\boxed{\epsilon' = L \epsilon L^T} \quad \text{--- (23)}$$

This proves that Strain is a 2nd order tensor.

Note:- Since Strain is a Symmetric 2nd order tensor, all theorems related to

- principal Stresses
- principal Stress direction
- max Shear Stress etc.

Apply directly to Strains.

① The directions of the principal normal strains are defined by

$$\tan 2\theta = \frac{2\epsilon_{xy}}{\epsilon_{xx} - \epsilon_{yy}} \quad \text{--- (24)}$$

& the two principal strains ϵ_1 & ϵ_2 are given by

$$\epsilon_1, \epsilon_2 = \frac{1}{2}(\epsilon_{xx} + \epsilon_{yy}) \pm \sqrt{(\epsilon_{xy})^2 + \frac{1}{4}(\epsilon_{xx} - \epsilon_{yy})^2} \quad \text{--- (25)}$$

② Relative to principal axes, the normal and Shear Strain in some direction rotated by an angle θ from the direction of max normal strain which in this context will be.

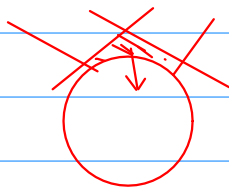
$$\epsilon = \epsilon_1 \cos^2 \theta + \epsilon_2 \sin^2 \theta \quad \text{--- (26)}$$

$$\gamma = -\frac{1}{2}(\epsilon_1 - \epsilon_2) \sin 2\theta.$$

(27)

⇒ Mohr Constructions → Similar to Stress.

③ Strain tensor quantifies → Stretching / distortion undergone by rock.



Other types of deformation that do not lead to any strain.

↓
Rigid body translation. (All particles displaced by same amount)
 $\vec{x}^* = \vec{x} - \vec{a}$

In 2D, rigid body translation is

$$x^* = x - a, \quad y^* = y - b \quad \text{--- (28)}$$

∴ a & b are components of a constant vector.

∴ $\boxed{\nabla \vec{u} = 0}$ for translatory motion.

Infinitesimal Rigid Body Rotations

From Eq (17)

$$\epsilon = \text{Sym}(\nabla \vec{u}) = \frac{1}{2} [\nabla \vec{u} + (\nabla \vec{u})^T]$$

Statement:-

Any deformation whose displacement gradient is anti-symmetric will lead to no infinitesimal strain.

Proof:-

consider rigid body rotation of particle about 'O'

This is mathematically similar but different from rotation of co-ordinate system

$$\begin{aligned} x^* &= r \cos(\theta + \phi) = r \cos \theta \cos \phi - r \sin \theta \sin \phi \\ y^* &= r \sin(\theta + \phi) = r \sin \theta \cos \phi + r \cos \theta \sin \phi \end{aligned}$$

$$\text{or, } x^* = x \cos \phi - y \sin \phi, y^* = x \sin \phi + y \cos \phi \quad \text{--- (29)}$$

Now, $\therefore x^* = x - u$

$\therefore$ displacement components

$$u = x - x^* = x(1 - \cos \phi) + y \sin \phi \quad \text{--- (30)}$$

$$v = y - y^* = -x \sin \phi + y(1 - \cos \phi) \quad \text{--- (31)}$$

So displacement gradient-

$$\nabla u = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix} = \begin{bmatrix} 1 - \cos \phi & \sin \phi \\ -\sin \phi & 1 - \cos \phi \end{bmatrix} \quad \text{--- (32)}$$

Since the rotation is infinitesimal -

Expand trigonometric terms in Taylor's Series and ignoring higher than the order in ϕ

$$\nabla u = \begin{bmatrix} 0 & \phi \\ -\phi & 0 \end{bmatrix} \quad \text{--- (33)}$$

$\therefore$ The displacement gradient corresponding to infinitesimal rigid body is Anti-symmetric. $\Rightarrow (\nabla \vec{u})^T = -(\nabla \vec{u})$

This type of rotation leads to no-strain

Since
$$\epsilon = \frac{1}{2} [\nabla u + (\nabla u)^T] = 0$$

In general we can define the rotation tensor ω as anti-symmetric part of displacement gradient

Note the difference w.r.t Co-ordinate transf

$$\omega = \text{asym}(\nabla u) = \frac{1}{2} [\nabla u - (\nabla u)^T]$$

$$= \begin{bmatrix} 0 & \frac{1}{2}(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x}) \\ -\frac{1}{2}(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x}) & 0 \end{bmatrix} \quad \text{--- (34)}$$

$\Rightarrow$ from eqn (17) & (34)

$$\boxed{\nabla u = \epsilon + \omega}$$

Conclusions

Infinitesimal deformations can always be thought of being composed of 3-components

- ($\nabla \vec{u}$)
- (i) ✓ Rigid body translations (which gives rise to no strain or rotation).
 - (ii) ✓ Stretching and/or distorting deformations, gives rise to a non-zero strain tensor and
 - (iii) ✓ Infinitesimal rotation, which gives rise to a non-zero rotation tensor.